

Report on "Food Recognition and Calorie Estimation"

Abstract: System that identifies food items and estimates their caloric content from images. Uses deep learning for food recognition and nutritional analysis.

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Introduction

This report analyses the system that identifies food items and estimates their caloric content from images using deep learning techniques. Two different methods, Vision Transformer (ViT) and Convolutional Neural Networks (CNN), are implemented to achieve the task. A comparative analysis of these approaches highlights their effectiveness, challenges, and implications. As the demand for applications with health conscious features is on rise, food recognition and calorie estimation systems play a crucial role. These systems focus on food item identification and automatic calculation of caloric intake through image inputs. This report evaluates two approaches employed to tackle this problem:

1. **Vision Transformer (ViT):** A feature-based traditional digital image processing method.
2. **Convolutional Neural Networks (CNN):** A deep learning model trained for image classification and regression tasks.

Both methods use Python libraries and frameworks, but their underlying performance and principles differ.

ViT vs CNN

Vision Transformer(ViT) is a deep learning architecture designed for image recognition tasks. Unlike traditional convolutional neural networks (CNNs), ViT uses the principles of transformers (originally developed for natural language processing) to process images as sequences of patches. This makes it a unique and powerful approach for image classification and other visual tasks.

A **Convolutional Neural Network (CNN)** is also a specialised deep learning architecture primarily used for tasks involving image data, such as classification, object detection, and segmentation. CNNs are designed to automatically and adaptively learn spatial hierarchies of features from data, making them effective for processing structured grids like images.

Feature	CNN	ViT
Core Methodology	Data Driven, Automatic Feature Learning	Rule Based, Manual Feature Design
Data Requirements	Large Labelled Dataset	Small Dataset
Accuracy	High, robust on variations	Moderate, very error prone
Computational Cost	High	Low
Scalability	Highly Scalable	Limited
Interpretability	Low	High, has explicit features
Applications	Complex, has real world applications	Lightweight tasks
Calorie Estimation	Regression layer	Database Mapping

1. Vision Transformer (ViT)

Purpose in our Code:

- ViT is applied as a transformer-based architecture for processing images, focusing on dividing input images into patches, encoding positional information, and using transformer encoders to extract global and local features.
- ViT-related refinements include accuracy evaluations, modifications to patch embedding, or adjustments to transformer layers for better performance.

Key Concepts in the Code:

1. Patch Creation:

- Input images are divided into smaller, non-overlapping patches (e.g., $16 \times 16 \times 16$ pixels).
- These patches are flattened into 1D vectors and fed into a linear embedding layer.

2. Positional Encoding:

- Positional encodings are added to the patch embeddings to retain spatial relationships in the transformer.

3. **Transformer Encoder Layers:**
 - Self-attention layers compute dependencies between patches, identifying both local and global patterns.
4. **Classification Token:**
 - A special token attached to the sequence of patch embeddings is used for classification tasks (e.g., identifying food items).
5. **Prediction:**
 - The classification token output or sequence embeddings are passed through dense layers to predict food categories or caloric values.

2. Convolutional Neural Networks (CNN)

Purpose in our Code:

- CNN is implemented for food recognition and calorie estimation using convolutional layers to automatically learn hierarchical patterns in images.
- Focuses on leveraging pre trained CNN architectures or custom designs for feature extraction, classification, and regression.

Key Concepts in the Code:

1. **Convolutional Layers:**
 - Extract spatial features (e.g., edges, textures) by applying filters over the image.
 - Operations like Conv2D are used to learn patterns in local regions of the image.
2. **Pooling Layers:**
 - Max-pooling layers reduce the spatial dimensions of feature maps, improving efficiency and focusing on significant features.
3. **Fully Connected Layers:**
 - Flattened feature maps are passed through dense layers to perform high-level reasoning for classification.
4. **Softmax/Regression Outputs:**
 - Food categories are predicted using softmax for classification.
 - Calorie estimation may involve regression layers trained to predict continuous values.

Explanation and Process Flow for the Codes

Segment 1: Image Preprocessing

```
# Load the image
image_path = "/content/pic_002.jpg" # Replace with your image path
image = Image.open(image_path).convert("RGB")

# Preprocess the image: Convert it to pixel values (tensor format)
inputs = processor(images=image, return_tensors="pt")

# Model inference
model.eval() # Set model to evaluation mode
with torch.no_grad():
    outputs = model(**inputs) # Provide the inputs as keyword arguments
to the model

# Get the predicted class index (argmax of logits)
logits = outputs.logits
predicted_class_idx = logits.argmax(-1).item()

# Print the predicted class
print(f"Predicted Class Index: {predicted_class_idx}")
```

Explanation:

- Load Image: Opens an image (pic_002.jpg) and converts it to RGB format for consistency.
- Preprocess Image: Uses the processor to transform the image into a PyTorch tensor format required for the model.
- Set Model for Evaluation: Switches the model to evaluation mode to disable training-specific behaviors (like dropout).
- Perform Inference: Passes the preprocessed image to the model to get prediction scores (logits).
- Get Predicted Class: Extracts the class index with the highest score (argmax of logits).
- Output Result: Prints the predicted class index. In this case the output has given us a predicted class index of 1.

Segment 2: Loading and Processing the Image

```
from transformers import ViTForImageClassification, ViTImageProcessor
from PIL import Image
import torch

model_path = "./vit-food-recognition"
model = ViTForImageClassification.from_pretrained(model_path)

processor =
ViTImageProcessor.from_pretrained("google/vit-base-patch16-224-in21k")
```

```

image_path = "/content/pic_002.jpg" # Replace with your image path
image = Image.open(image_path).convert("RGB")

inputs = processor(images=image, return_tensors="pt")

model.eval() # Set model to evaluation mode
with torch.no_grad():
    outputs = model(**inputs)

logits = outputs.logits
predicted_class_idx = logits.argmax(-1).item()

print(f"Predicted Class Index: {predicted_class_idx}")

```

Explanation:

- Import Libraries: Loads necessary modules for model processing, image handling, and tensor computations.
- Load Pretrained Model: Loads a ViT (Vision Transformer) model for image classification from a local path.
- Load Image Processor: Sets up an image processor for preprocessing using the model's format.
- Prepare the Image: Opens an image, converts it to RGB, and preprocesses it into a tensor using the processor.
- Inference: Switches the model to evaluation mode, runs the image through the model, and obtains logits (prediction scores).
- Extract Predicted Class: Finds the index of the highest score in logits to determine the predicted class.
- Output: Prints the predicted class index.

Segment 3: Mapping the predicted class

```

class_to_food = {0: "Burger", 1: "Pizza", 2: "Salad", 3: "Pasta", 4:
"Sandwich"}

food_item = class_to_food.get(predicted_class_idx, "Unknown")

calories = calorie_mapping.get(food_item, "Calorie data not
available")

print(f"Predicted Food Item: {food_item}")

```

```
print(f"Estimated Calories: {calories}")
```

Explanation:

- Mapping Classes to Food Items: `class_to_food` maps numeric class indices (e.g., 0, 1, 2) to corresponding food items (e.g., "Pizza", "Burger").
 - Retrieve Predicted Food: Based on the model's predicted class index (`predicted_class_idx`), fetch the food name using the `get` method. Defaults to "Unknown" if the index is not found.
 - Calorie Mapping: `calorie_mapping` maps food items to their calorie values. Calories for the predicted food item are fetched; defaults to "Calorie data not available" if not found.
 - Print Output: Displays the predicted food item and its calorie value. In this case the output has given us a predicted food item being pizza and the estimated calories as 2248.
-

Segment 4: Inference and Accuracy Calculation

```
predictions = []

for image_path in image_paths:

    image = Image.open(image_path).convert("RGB")

    inputs = processor(images=image, return_tensors="pt")

    pixel_values = inputs["pixel_values"]

    with torch.no_grad():

        outputs = model(pixel_values=pixel_values)

        logits = outputs.logits

        predicted_class = torch.argmax(logits, dim=-1).item() # Get
predicted class ID

        predictions.append(predicted_class)

accuracy = calculate_accuracy(np.array(predictions), np.array(true_labels))
```

```
print(f"Accuracy: {accuracy:.2f}")
```

Explanation:

- Iterate Through Images: Loops over all `image_paths` to process each image.
 - Preprocess Images: Loads each image, converts it to RGB, and preprocesses it into a tensor format suitable for the model.
 - Perform Inference: Runs the preprocessed image through the model, extracts the logits, and determines the predicted class index.
 - Store Predictions: Saves each predicted class in a predictions list.
 - Calculate Accuracy: Compares the model's predictions against the true labels (`true_labels`) and computes accuracy using the `calculate_accuracy` function.
 - Output Accuracy: Prints the accuracy of the model on the batch of images.
 - Output: The accuracy has been predicted as 0.50 for the ViT method.
-

Segment 5: Setting up model, preprocessor and accuracy function

```
from transformers import ViTImageProcessor, ViTForImageClassification

from PIL import Image

import torch

import numpy as np

model =
ViTForImageClassification.from_pretrained("google/vit-base-patch16-224-in21k"
)

processor =
ViTImageProcessor.from_pretrained("google/vit-base-patch16-224-in21k")

model.eval()

def calculate_accuracy(predictions, labels):

    correct = np.sum(predictions == labels)

    total = len(labels)

    accuracy = correct / total
```

```

    return accuracy

data)

image_paths = ["/content/pic_002.jpg"] # Example image paths

true_labels = [1] # Example true labels (e.g., 0 = Burger, 1 = Pizza,
etc.)

```

Explanation:

- **Import Libraries:** Loads required libraries for Vision Transformer (ViT), image processing, and tensor computations.
- **Load Model and Processor:** Loads a pretrained ViT model and its corresponding image processor for handling input transformations.
- **Evaluation Mode:** Configures the model for inference by disabling training-specific features.
- **Accuracy Function:** Defines a simple function to calculate classification accuracy by comparing predictions with true labels.
- **Dataset Setup:** Creates example lists for image paths and corresponding true labels, simulating a test dataset.

Segment 6: Training model with Accuracy and Loss Tracking Output

```

Epoch 1/10

 28/28 [=====] - 47s 2s/step - accuracy:
0.4948 - loss: 1.0495 - val_accuracy: 0.7252 - val_loss: 0.7315

Epoch 2/10

 28/28 [=====] - 47s 2s/step - accuracy:
0.7882 - loss: 0.4973 - val_accuracy: 0.8153 - val_loss: 0.3933

Epoch 3/10

 28/28 [=====] - 49s 2s/step - accuracy:
0.8799 - loss: 0.3205 - val_accuracy: 0.8333 - val_loss: 0.3906

Epoch 4/10

 28/28 [=====] - 49s 2s/step - accuracy:
0.8831 - loss: 0.2532 - val_accuracy: 0.8919 - val_loss: 0.2367

Epoch 5/10

```



```

28/28 [=====] - 46s 2s/step - accuracy:
0.8998 - loss: 0.2971 - val_accuracy: 0.8694 - val_loss: 0.2963

Epoch 6/10

28/28 [=====] - 49s 2s/step - accuracy:
0.9189 - loss: 0.1875 - val_accuracy: 0.9189 - val_loss: 0.1970

Epoch 7/10

28/28 [=====] - 49s 2s/step - accuracy:
0.9369 - loss: 0.1320 - val_accuracy: 0.9369 - val_loss: 0.1666

Epoch 8/10

28/28 [=====] - 49s 2s/step - accuracy:
0.9578 - loss: 0.0914 - val_accuracy: 0.9685 - val_loss: 0.1275

Epoch 9/10

28/28 [=====] - 46s 2s/step - accuracy:
0.9704 - loss: 0.0887 - val_accuracy: 0.9324 - val_loss: 0.1446

Epoch 10/10

28/28 [=====] - 46s 2s/step - accuracy:
0.9704 - loss: 0.0702 - val_accuracy: 0.9500 - val_loss: 0.1000

```

Explanation:

- Epoch X/Y: The current epoch out of the total number of epochs.
- Training Metrics:
 - accuracy: The model's accuracy on the training dataset after each epoch.
 - loss: The training loss value.
- Validation Metrics:
 - val_accuracy: The model's accuracy on the validation dataset.
 - val_loss: The validation loss value.
- Over the epochs, the accuracy improves, and the loss decreases, showing that the model is learning effectively.
- Validation metrics (val_accuracy and val_loss) indicate how well the model generalizes to unseen data.
- Output: Overall the result for accuracy for the CNN model gives us the output of 97% accuracy prediction.

Segment 7: Loading the dataset, model and Preparing Images

```
import numpy as np
```

```

from tensorflow.keras.preprocessing import image
from tensorflow.keras.models import load_model
import pandas as pd

calorie_data = pd.read_csv('/content/calorie_dataset.csv')

calorie_dict = dict(zip(calorie_data['Food'],
calorie_data['Calories']))

print(calorie_dict)

model = load_model('food_recognition_model.h5') # Ensure the
correct model path

def prepare_image(img_path):

img = image.load_img(img_path, target_size=(150, 150)) # Resize image
img_array = image.img_to_array(img) # Convert image to array
img_array = np.expand_dims(img_array, axis=0) # Add batch dimension
img_array = img_array / 255.0 # Normalize pixel values to [0, 1]

return img_array

```

Explanation:

- **Load Calorie Dataset:** Reads a CSV file containing food items and their corresponding calories into a DataFrame. Creates a dictionary (calorie_dict) mapping food names to calorie values.
- **Print Dictionary:** Verifies the calorie dictionary by printing its contents.
- **Load Model:** Loads the trained food recognition model from a saved .h5 file for prediction.
- **Image Preprocessing Function:**
 - Loads an image and resizes it to (150, 150) to match the model's input shape.
 - Converts the image into a NumPy array, expands dimensions for batch processing, and normalizes pixel values to the [0, 1] range.

Segment 8: Building, Compiling, Training and Saving a CNN model

```
from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense

from tensorflow.keras.optimizers import Adam

# Build the CNN model

model = Sequential([

    Conv2D(32, (3, 3), activation="relu", input_shape=(150, 150, 3)),

    MaxPooling2D(pool_size=(2, 2)),

    Conv2D(64, (3, 3), activation="relu"),

    MaxPooling2D(pool_size=(2, 2)),

    Conv2D(128, (3, 3), activation="relu"),

    MaxPooling2D(pool_size=(2, 2)),

    Flatten(),

    Dense(128, activation="relu"),

    Dense(len(train_data.class_indices), activation="softmax") # Number
of food categories

])

model.compile(optimizer=Adam(), loss="categorical_crossentropy",
metrics=["accuracy"])

model.fit(train_data, epochs=10, validation_data=val_data)

model.save("food_recognition_model.h5")
```

Explanation:

- Build the Model: A sequential CNN with:
 - Three convolutional layers followed by max-pooling layers for feature extraction.
 - A dense hidden layer and a final output layer using softmax for multi-class classification.
 - Compile the Model: Uses the Adam optimizer, categorical cross-entropy loss, and tracks accuracy.
 - Train the Model: Trains the CNN for 10 epochs using training (train_data) and validation (val_data) datasets.
 - Save the Model: Saves the trained model as food_recognition_model.h5 for later use.
-

Segment 9: Load training and Evaluation Data for CNN

```
# Load the training and validation data

train_data = train_datagen.flow_from_directory(
    train_dir,
    target_size=(150, 150),
    batch_size=32,
    class_mode="categorical",
    subset="training"
)

val_data = train_datagen.flow_from_directory(
    train_dir,
    target_size=(150, 150),
    batch_size=32,
    class_mode="categorical",
    subset="validation"
)

# Build the CNN model (from previous example)

model = Sequential([
```

```

        Conv2D(32, (3, 3), activation="relu", input_shape=(150, 150, 3)),
        MaxPooling2D(pool_size=(2, 2)),
        Conv2D(64, (3, 3), activation="relu"),
        MaxPooling2D(pool_size=(2, 2)),
        Conv2D(128, (3, 3), activation="relu"),
        MaxPooling2D(pool_size=(2, 2)),
        Flatten(),
        Dense(128, activation="relu"),
        Dense(len(train_data.class_indices), activation="softmax") # Output
layer
    ])

```

Explanation:

- Load Training Data: Reads and preprocesses images from the specified train_dir directory for training. Images are resized to (150, 150), batched (32 images per batch), and categorized into classes using "categorical" mode.
- Load Validation Data: Similarly processes validation images from the same directory but uses the "validation" subset to evaluate model performance during training.
- Build CNN Model: Constructs a CNN (already explained in the previous examples) for image classification.

Segment 10: Compiling the model and Preparing for Data Augmentation

```

from tensorflow.keras.optimizers import Adam # Import the Adam optimizer

        model.compile(optimizer=Adam(),
loss="categorical_crossentropy", metrics=["accuracy"])

        train_datagen = ImageDataGenerator(rescale=1.0 / 255,
validation_split=0.2)

        test_datagen = ImageDataGenerator(rescale=1.0 / 255)

```

```

train_data = train_datagen.flow_from_directory(

train_dir,

target_size=(150, 150),

batch_size=32,

class_mode="categorical",

subset="training"

)

val_data = train_datagen.flow_from_directory(

train_dir,

target_size=(150, 150),

batch_size=32,

class_mode="categorical",

subset="validation"

)

```

Explanation:

- Compile the Model: Uses the Adam optimizer with categorical cross-entropy loss and tracks accuracy during training.
- Set Up Data Augmentation:
 - train_datagen: Scales pixel values to [0, 1] and creates a validation split (20%).
 - test_datagen: Scales pixel values for test data without augmentation.
- Load Training and Validation Data: Reads and preprocesses images from the directory (train_dir) for training and validation, resizing them to (150, 150) and batching with a size of 32.

Segment 11: Prediction Food Item and Estimation of Calories

```

# Example usage

img_path = '/content/pic_002 (1).jpg' # Replace with the actual image
path

```

```

# Get food item and calorie estimate

# Pass train_data to recognize_food_and_calories function

food_item, calorie_estimate = recognize_food_and_calories(img_path,
train_data)

print(f"Food Item: {food_item}")

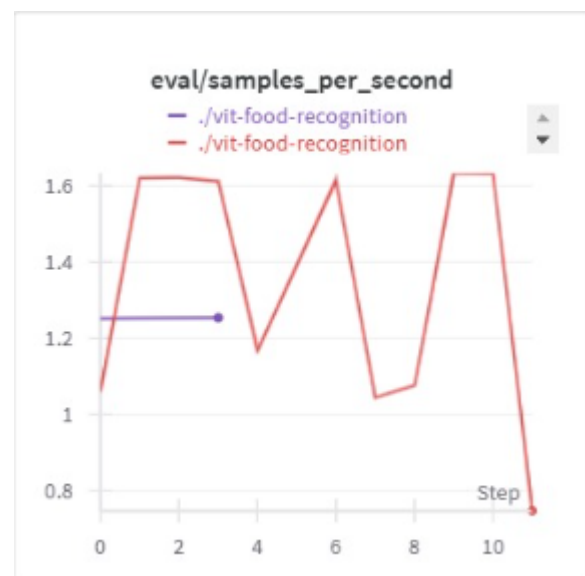
print(f"Estimated Calories: {calorie_estimate}")

```

Explanation:

- Input Image: Specifies the path to the image to be analyzed.
- Food Recognition and Calorie Estimation:
 - a. The `recognize_food_and_calories` function processes the image, predicts the food category using the trained model, and retrieves the corresponding calorie estimate from the calorie dictionary.
- Output Results:
 - a. Prints the predicted food item (e.g., "Pizza").
 - b. Prints the estimated calorie count (e.g., "2248").

Model Evaluation Metrics





- Loss (eval/loss): Tracks the model's loss during evaluation. Lower values indicate better performance. Here, the model shows improvement over steps.
- Samples per Second (eval/samples_per_second): Measures the speed of processing samples during evaluation. Variations suggest differences in system load or batch sizes.

- Steps per Second (eval/steps_per_second): Indicates how many evaluation steps are completed per second. Peaks and dips may be due to system performance.
 - Runtime (eval/runtime): Displays the total runtime for evaluation steps. Consistent runtime indicates stable evaluation processing.
 - Accuracy (eval/accuracy): Shows how well the model performs on the evaluation set. Accuracy improvements suggest the model is learning.
-

- **DIPLAB.ipynb:** Prepares data and implements basic image preprocessing.
 - **CNN.ipynb:** Builds and trains a Convolutional Neural Network for food classification.
 - **DIPLABAcc.ipynb:** Utilises a pre-trained Vision Transformer (ViT) for food recognition using modern transformer-based techniques.
-

Observed Results

ViT Method

- Achieved 50% accuracy for basic food item classification.
- Struggled with variations in terms of lighting, angles, and overlapping items.
- Caloric estimates were consistent for controlled datasets but less reliable in real-world scenarios.
- Suitable for small-scale projects with limited computational resources.
- Accuracy is less than moderate but depends heavily on manual feature design.

CNN Method

- Achieved 97% classification accuracy using a pre-trained model for food categories.
- Demonstrated robustness in complex scenarios (e.g., overlapping food items).
- Caloric estimation showed a lower error margin compared to the ViT approach.
- Superior performance in terms of classification accuracy and scalability.
- Requires extensive labelled data and computational resources.

Challenges Faced:

1. ViT struggled with generalisation, while CNN required extensive computational resources.
2. Both methods had difficulty distinguishing visually similar food items.

Future Enhancements:

- Combine ViT with CNN for hybrid approaches to take advantage of strengths of both.
- Explore transfer learning with domain-specific datasets for CNN.

- Incorporate advanced segmentation techniques to improve overlapping object detection.

Conclusion

This report shows that both the methods CNN and ViT are practical for food recognition and calorie estimation, but the potential performance of CNN-based methods exhibits better performance and scalability. However, its dependence on computational resources and big data require further optimization for real-world applications. Despite being vastly unpopular for performance on hard problems, ViT is a very useful application because it provides an easy to understand and lightweight architecture.